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AI-based synthetic data generation techniques for improved fault classification in power systems

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ABSTRACT

The unfaltering action of electricity networks depends on the early identification of faults. Several machine learning (ML) and deep learning (DL) techniques have been applied for fault detection and classification. However, their performance is highly reliant on the quantity and excellence of the dataset. This paper proposes AI-based synthetic data generation techniques that provide an excellent option for producing synthetic data. This synthetic data can then be used in combination with the original dataset, to generate a new dataset. This strategy involves augmenting the current dataset with synthetically created dataset points, resulting in a new dataset. This new dataset leads to more robust and efficient fault identification and classification using either machine or deep learning techniques. The techniques train on a collection of actual datasets and produce an artificial dataset that mimics the features of the actual dataset. The results show that the new dataset with the original and synthetic dataset combined shows better fault classification results. It is also observed that the performance of the ML algorithms is reliant on the method used for creating the dataset. GAN and VAE-GAN emerged as the most adjustable methods, which enhanced the generalizability of different models. It was found that the Decision Tree and Gaussian Naïve Bayes models are the most effective, with near-perfect metrics across the majority of synthetic data generation techniques implemented in this paper.

1. Introduction

Power systems are vital infrastructures that transport energy from the generating stations to the areas of utility. The continuously increasing electricity consumption leads to complex networks. Abnormal conditions can arise from a variety of causes, including environmental, unintentional, incidental, and aging factors. Any aberrant situation might harm both the generation and utilization. So, this failure investigation is an important topic of research [1,2]. The identification and localization of defects in power transmission networks is critical, and sophisticated signal processing techniques for this purpose are becoming increasingly common. Moreover, due to the dynamic behaviour of electrical power systems, these require modern optimization methods to accomplish a solution that fulfils the problem conditions, for example, the metaheuristic methods, ML algorithms, and many other techniques. Metaheuristic algorithms are mainly inspired by natural phenomena, animal behaviours, and biological processes [3–6]. Machine learning algorithms are mainly motivated by the way humans learn and process information. ML algorithms are trained using arithmetical numbers and scientific models to detect fault patterns with little human involvement [7].

Unfortunately, getting labelled data presents substantial problems and time limits, especially in power systems with uncharacteristic behaviours that are uncommon and frequently unforeseen. The following are some of the challenges faced in the power systems:

1. Sensor or other monitoring equipment failure may lead to missing data.
2. Infrequent fault occurrences may lead to sparse datasets.
3. As fault occurrences are rare, there is a possibility of imbalanced datasets.
4. The data collected can be noisy due to environmental factors.
5. Power systems create complex data that can be difficult to manage. It might also be difficult to process the data and extract meaningful information from it.
6. The dynamic nature of the power system makes the adaptation of the trained models difficult.

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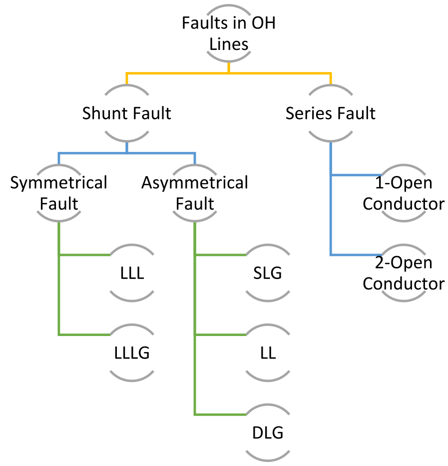

Fig. 1. Faults classification considered.

To overcome this issue, current research has looked at the feasibility of using synthetically generated data to improve the accuracy of ML models. Specifically, Generative Adversarial Networks (GAN) and Variational Encoders (VAE) have been used to generate a fake dataset that meticulously resembles the unique data distribution [8,9]. VAEs are data formation architectures that may be trained as a reduced-dimensional depiction of the input dataset and then used to produce a novel dataset. A signal spectrum-based ML strategy was used for a variety of algorithms to detect hidden patterns of problematic situations through predictive maintenance [10].

An acoustic emission-based problem diagnostic method for power transformers was proposed in [11]. A VAE-based technique that augments the restricted labelled data and outperforms existing machine learning methods was suggested in [12]. A unique double-circuit transmission line protection technology that uses KNN to identify and correctly localize shunt faults was implemented in [13]. A technique for defect diagnosis in wind turbines using VAE and synthetic data was proposed in [14].

After exploring the existing literature in this domain, the following gaps were identified –

1. Mostly VAE and GAN [8,9,11,14] were discussed. Using these limited methods is not feasible as it is possible that all the intricate details of the dataset may not be captured by these methods. So, there is a need to explore other methods for data generation.
2. As only VAE [8] is applied for generating synthetic data for fault classification in power systems, exploring other methods for this specific domain is important.
3. There is a lack of comparative works for estimating the results produced by different methods.
4. Mostly, the synthetically generated dataset is investigated using the KNN algorithm [13]. But a wider exploration is required.

So, the key contributions of this paper are –

1. This study studies various synthetic data generation (syn-gen) methods. Firstly, SMOTE (Synthetic Minority Over-sampling Technique), a widely used method for creating syn-gen data in feature space, would be implemented. Next, a deep learning-based method, GAN, that uses two contending neural networks (generator and discriminator) to produce syn-gen would be applied. Next, a numerical method that generates data by estimating the probability density function of the real dataset, Kernel Density Estimation (KDE) would be implemented.
2. Additionally, two hybrid syn-gen methods such as VAE-GAN and Principal Component Analysis with Gaussian Mixture Model (PCA-GMM) will also be explored. VAE-GAN uses latent space learning to generate high-quality samples to produce a syn-gen. In PCA-GMM, PCA helps in dimensionality reduction and GMM models data distribution.
3. The purpose is to evaluate the relevance and efficacy of these techniques for creating synthetic datasets in power systems and other applications.
4. A comparative analysis of the aforementioned syn-gen methods will be presented.
5. The paper will test and analyze numerous machine learning algorithms to see how syn-gen data affects their performance. Various ML techniques implemented are
   a. LR (Logistic Regression)
   b. ABC (AdaBoost Classifier)
   c. DT (Decision Tree)
   d. SVC (Support Vector Classifier)
   e. MLP (Multi-Layer Perceptron)
   f. GNB (Gaussian Naive Bayes)
6. This comparison will evaluate the impact of synthetic data on various ML models and check their sturdiness and performance. The influence of synthetic data on these models will be measured using common measures, including accuracy, precision, recall, F1-score, and AUC-ROC. The analysis will provide insights on how the syn-gen data helped in performance improvement. This comparison will give useful information on the efficiency of synthetic data in improving the robustness and generalisability of machine learning models.

To achieve the above-stated contributions, the paper has the following objectives:

1. To better comprehend theoretical topics, use statistical syn-generation approaches.
2. To implement hybrid syn-gen approaches for high-quality datasets.
3. To estimate the efficiency of the syn-gen approaches.
4. To assess the efficacy of original and augmented (original + syn-gen) datasets using different machine learning models.
5. To infer the effect of syn-gen on performance indicators.

The fault classification considered in this paper is (Fig. 1):

A. Shunt (Short Circuit) Fault
   a. Symmetrical Fault
      i. Triple Line
      ii. Triple Line-to-Ground
   b. Asymmetrical Fault
      i. Single Line-to-Ground
      ii. Double Line
      iii. Double Line-to-Ground
B. Series (Open Circuit)
a. 1- Open Conductor Fault
b. 2- Open Conductor Fault

2. Synthetic data generation

The time-consuming and costly nature of data collection and annotation [15] leads to several problems. Because machine learning relies so largely on data, some of the main obstacles and difficulties it encounters are as follows:

- One of the biggest issues facing machine learning experts is ensuring the quality of the data. Models may produce inaccurate or vague estimates owing to misperception and delusion when the dataset quality is poor [16 17].

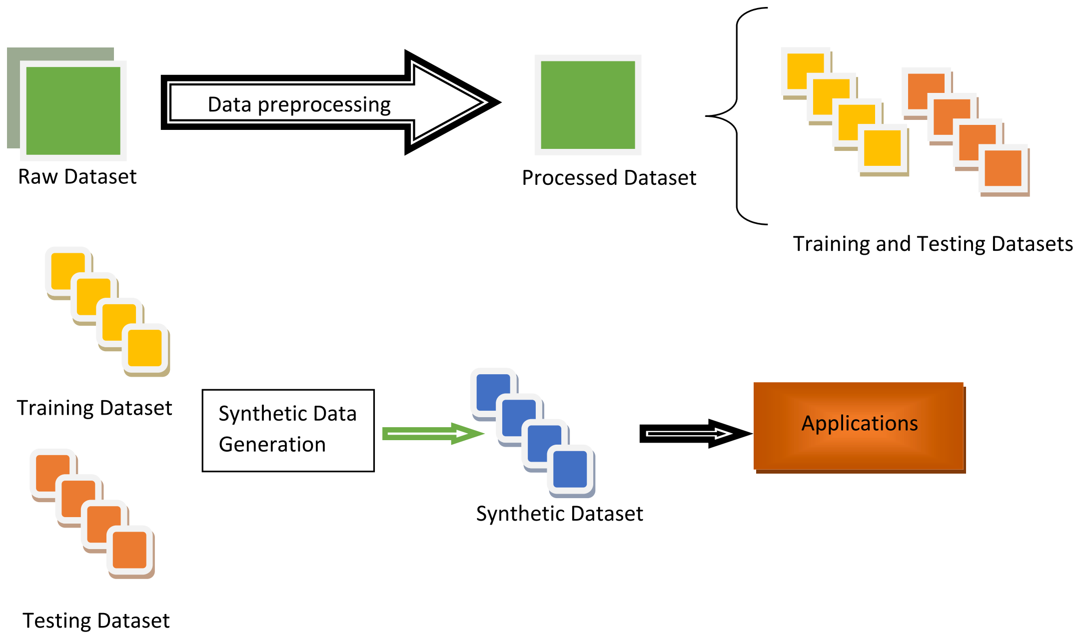

Fig. 2. Synthetic data generation workflow.

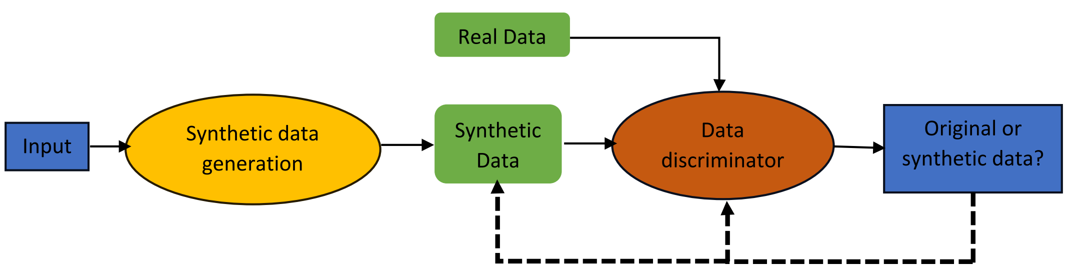

Fig. 3. GAN.

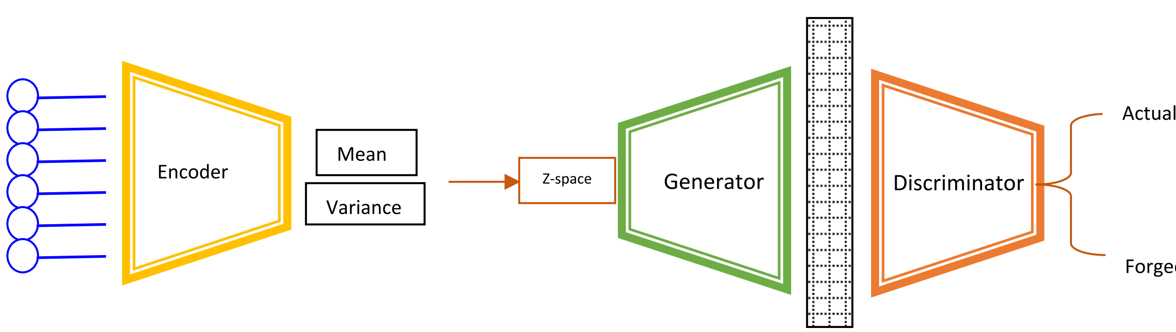

Fig. 4. VAE-GAN.

- A large fraction of the current AI problem is caused by a lack of data availability: either there are not enough datasets available, or labelling data by hand is too expensive [18].
- There are several situations when privacy and fairness concerns prevent datasets from being made publicly available.

Syn-gen (Fig. 2) may be quite helpful in these situations, and we will look into methods for producing anonymized datasets with varying degrees of privacy protection. For creating the synthetic data, the training datasets are used to understand the patterns, data types, statistical properties, and anomalies of the dataset. Once this is understood, then various statistical and ML methods can be implemented for syn-gen, depending on the data types and complexities of the dataset. The next step is the validation of the syn-gen data by testing the trained models on it. After the syn-gen dataset is validated, it can be augmented in the original dataset and deployed for various applications.

Deep learning-based syn-gen models employ unsupervised learning to understand how data is distributed and the trends in it, and have drawn considerable interest lately. The syn-gen methods implemented in this paper are (Fig. 3) –

1. SMOTE (Synthetic Minority Oversampling Technique)
2. GAN (Generative Adversarial Network)

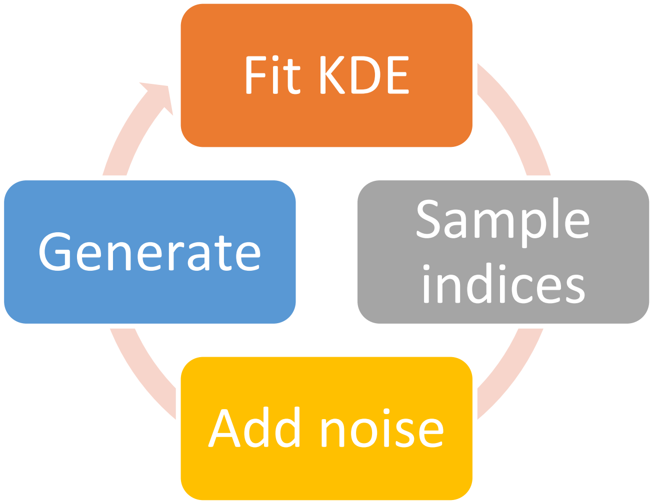

Fig. 5. KDE.

3. VAE-GAN (Variational Autoencoder Generative Adversarial Network)
4. PCA-GMM (Principal Component Analysis-Gaussian Mixture Model)
5. KDE (Kernel Density Estimation)

1. Smote

It is an oversampling strategy for syn-gen by a perturbation approach [19]. Assume a data value and its neighbour from a less-represented group, and compute the linear gap between them. This gap is scaled by an arbitrary value between 0 and 1, then augmented to the original data value. As a result, the novel data value is positioned along the linear gap connecting the initial data value and its neighbour. This procedure is iterated till the anticipated data values are attained.

Consider a given smaller class occurrence a_i with n nearest neighbour in space – a_1, a_2, … a_n.

Generate synthetic data by selecting a random dataset point, a_j, from the n neighbours using the equation –

$$a_{new} = a_i + \alpha(a_j - a_i) \tag{1}$$

where, a_i – original smaller class occurrence; a_j – random dataset point; α – random number with uniform distribution between 0 and 1.

2. GAN

GAN is based on an argumentative process of generating approximate DL models [20]. It has two models [21] (Fig. 3): one for generation and the other for discrimination. The first model produces synthetic data based on the initial data, but it is different from it. The second model distinguishes between the original and synthetic data. An argumentative process between the two models continues as the first model tries to overcome the second model to create artificial data resembling the original data.

3. VAE-GAN

It [22] (Fig. 4) has an encoder module that has mean and variance as the input by minimizing the prior values. This is recreated in z-space to have minimum MSE and then given to the generator. If the MSE is not kept to a minimum, the created sequence is not considered. The discriminator tries to identify and classify the input and the recreated sequence as actual and forged.

4. PCA-GMM

PCA and GMM were combined to develop a hybrid model for syn-gen in [23]. PCA is used for reducing the dimensions of the dataset. Consider a given dataset D with x samples and f features –

Calculate the average for each feature and center the data using the following equation –

$$D_{centered} = D - \sigma \tag{2}$$

where σ is the average vector.

Calculate the covariance using the equation –

$$Cov = \frac{1}{x-1} D^T_{centered} D_{centered} \tag{3}$$

If E is a vector of the eigenvalues, then principal components are calculated using the following equation –

$$D_{PCA} = D_{centered} E_k \tag{4}$$

where E_k contains k top eigenvectors. After dimensionality reduction, GMM is used to model the Gaussian distribution of w components of the reduced dataset using the equation –

$$PDF = \sum_{i=1}^{w} wt_i GD \tag{5}$$

where PDF is the probability density function, wt is weights, and GD is the Gaussian distribution function. To generate the synthetic data, sample a dataset point from the GD and reconstruct the dataset is needed by projecting it back from the PCA.

5. KDE

It is a non-parametric method. It is based on the PDF of the dataset. The new samples are generated by following the estimated distribution of the original dataset. The following process (Fig. 5) is reiterated to synthetically generate the dataset:

1. Fit KDE
2. Sample indices
3. Add noise
4. Generate a synthetic dataset

For random sample ($s = s_1, s_2, \ldots, s_m$) with unknown density function with bw as the bandwidth parameter and kf the kernel function is given as [24]:

$$f(s) = \frac{1}{m} \sum_{i=1}^{m} \frac{1}{bw} kf\left(\frac{s - s_i}{bw}\right) \tag{6}$$

After fitting the KDE, the following phase is to select samples from the predicted distribution. Syn-gen data is created by selecting random indices from the original dataset and adding noise to them. This data is used for ML training. Cross-validation is used to improve bw, which impacts the syn-gen data quality. For this, several bandwidth settings are tested, and the one with the lower error or increased performance metric is selected.

3. Proposed methodology

A large dataset is required to create effective ML models. But when the real-life dataset is limited, syn-gen models provide a lucrative option to create more datasets. ML and DL algorithms can help to generate synthetic data from existing datasets. The following is the workflow (Fig. 6) for the proposed methodology –.

1. Imports and Setup:
   - o Import necessary libraries.
   - o Define input and latent space dimensions.
2. Model Definitions:
   - o Encoder: Defines the encoder model with input, hidden, and latent layers.
   - o Sampling Function: Defines how to sample from the latent space using a reparameterization trick.
     - o Decoder: Defines the decoder model.
     - o Generator: Uses the same architecture as the decoder.

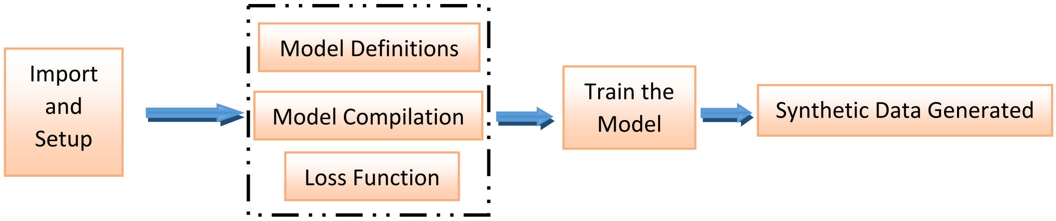

Fig. 6. Proposed methodology.

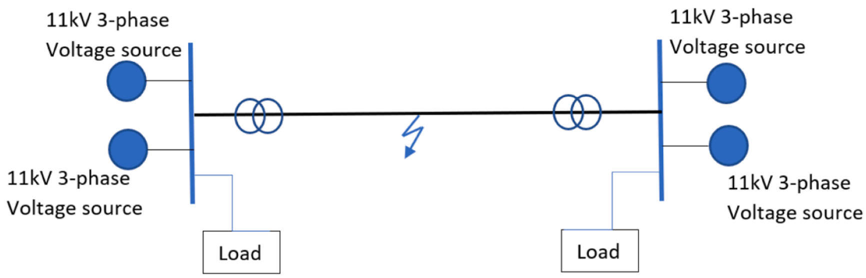

Fig. 7. Test system.

o Discriminator: Defines the discriminator model.

3. Model Compilation:
o Build encoder, decoder, generator, and discriminator models.
o Compile the discriminator model.
o Define the model with inputs, encoder, decoder, and discriminator.
4. Loss Functions:
o Define reconstruction loss.
o Add and compile combined loss.
5. Training Loop:
o Train the discriminator with real and fake samples.
o Train the generator via the model.
o Print progress at intervals.
6. Generating Synthetic Data:
o Use the trained generator to produce synthetic data.
o Convert the synthetic data into a DataFrame and save it as a CSV file.

The Model Definitions, Compilation, and Loss Functions depend on the syn-gen technique being implemented.

The pseudocode for syn-gen is as follows:

#Imports and setup

Start
Import required_libraries
Set input_features, latent_space, epochs, batch_size, learning_rate, decoder, encoder
End

#Model Components

Start
Encoder
Input data_sample of input_dimension

Pass through transformation layers

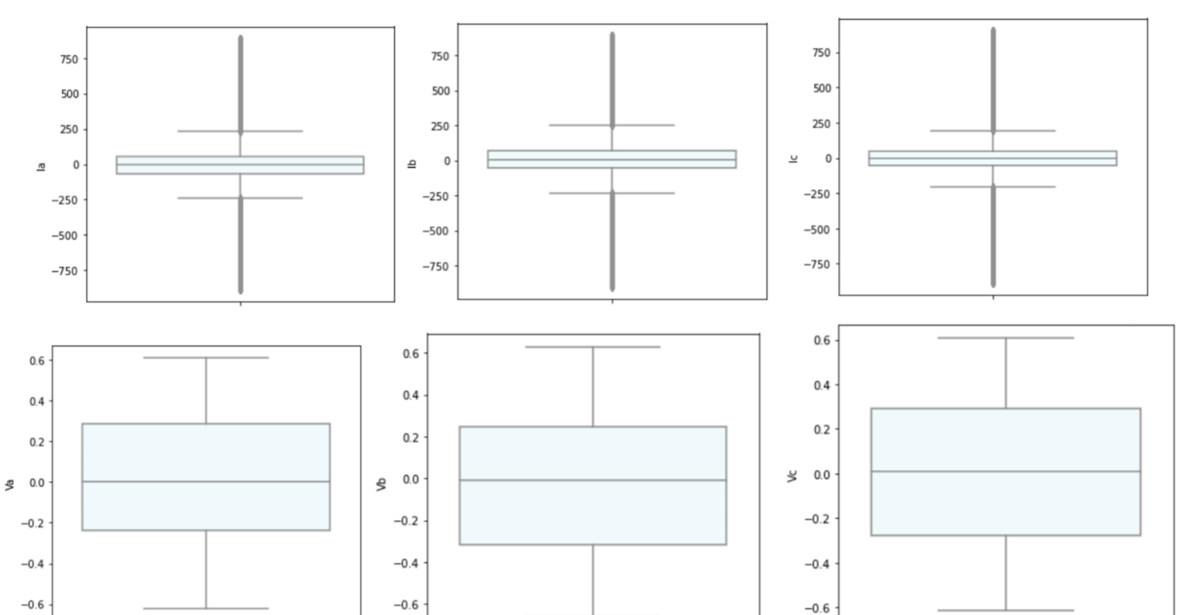

Fig. 8. Test system dataset.

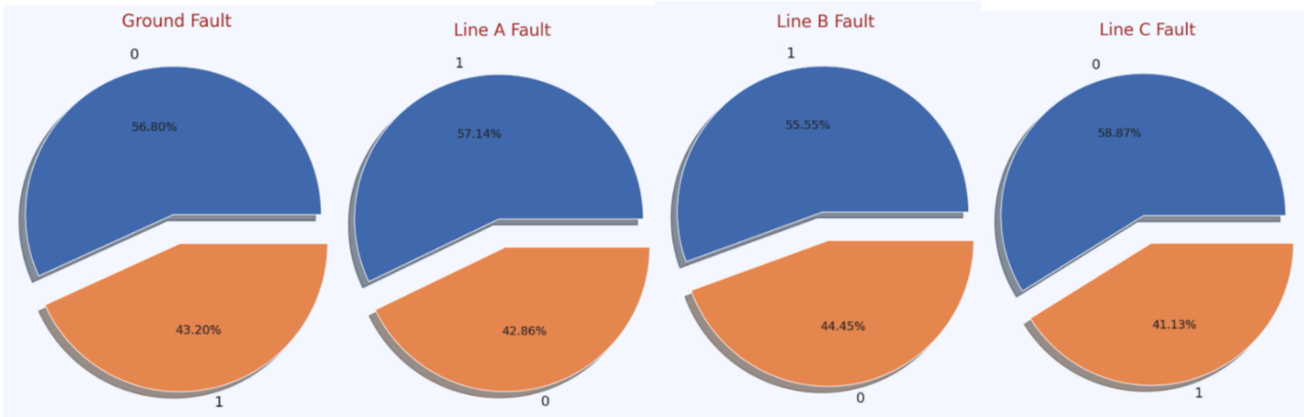

Fig. 9. Fault occurrences on lines G, A, B, and C.

Table 1
Fault details.

Fault Type	Fault Type Interpretation	Number of Fault Occurrences
0000	No fault	2365
1011	LLG fault	1134
1111	LLLG fault	1133
1001	LG fault	1129
01110110	LLL faultLL fault	10961004

Output encoded_sample

Decoder
Input encoded_sample

Pass through reconstruction layers

Output reconstructed_sample
End

#Compile models

Configure encoder
Configure decoder

#Training loop

Encode data
Decode data

4. Test system and data preprocessing

The test system is a simulation model used to identify and categorize electrical faults. Before executing the simulation model, it is crucial to define a parameter for every electrical block. Throughout this simulation, the voltage and current amplitudes will be monitored. The current amplitude would be significantly increased in the event of a breakdown or grid power loss. However, the voltage amplitude will decrease as a result of transmission line energy losses. Four three-phase voltage sources are shown in the test system (Fig. 7). Each source has an 11 kV rating. At the end of the transmission line, two sources are joined via transformers. The transmission line's midpoint is used to mimic the failure. Next, both fault and normal situations were simulated for the test system. Line-to-line, line-to-ground, double line-to-ground, triple line, and triple line-to-ground short circuit faults were all simulated, and the resulting dataset was recorded and used for syn-gen.

According to the test system, the dataset (Fig. 8) included four output features (G, A, B, and C) that represented the fault position in the respective lines, three input attributes (Va, Vb, and Vc), and three currents (Ia, Ib, and Ic). For both normal and fault circumstances, the four output properties had Boolean values of 0 and 1, respectively. The fault's label, FaultType, was determined by these values.

	G	C	B	A	Ia	Ib	Ic	Va	Vb	Vc	FaultType
0	1	0	0	1	-151.291812	-9.677452	85.800162	0.400750	-0.132935	-0.267815	1001
1	1	0	0	1	-336.186183	-76.283262	18.328897	0.312732	-0.123633	-0.189099	1001
2	1	0	0	1	-502.891583	-174.648023	-80.924663	0.265728	-0.114301	-0.151428	1001
3	1	0	0	1	-593.941905	-217.703359	-124.891924	0.235511	-0.104940	-0.130570	1001
4	1	0	0	1	-643.663617	-224.159427	-132.282815	0.209537	-0.095554	-0.113983	1001

Compute loss
Update encoder and decoder weights
Log training progress

#After training

Create random encoded data
Decode_created_data
Obtain_syn-gen_data
Transform to usable format

A crucial preprocessing step in changing object-type fields to numerical-type is label encoding. Label encoding was done using the sklearn.preprocessing library package in Python 3.0. As seen by the boxplots of the voltages and currents, the dataset investigation revealed that it was almost regularly distributed and free of outliers.

Six output labels—no fault, LG fault, LL fault, DLG fault, LLL fault, and TLG fault—were included in the dataset. Fig. 9 displays the fault occurrences on lines G, A, B, and C. It has been noted that the number of fault occurrences is almost the same for all lines, making the dataset a

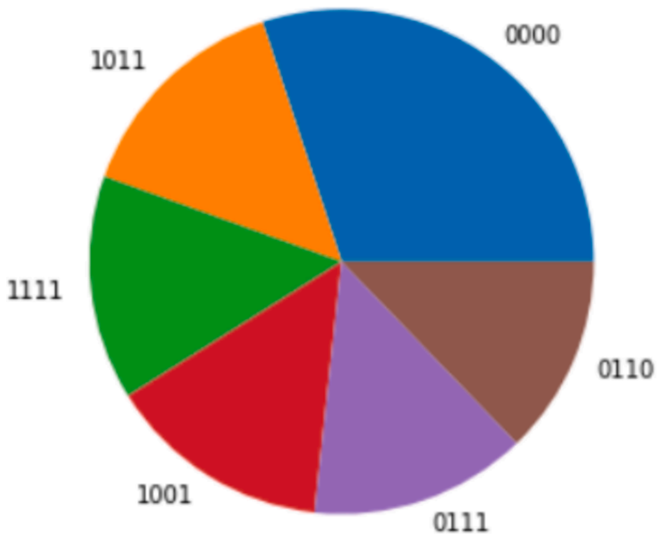

Fig. 10. Number of faults occurrences.

balanced dataset.

Table 1 displays the labels for the various faults. Fig. 10 displays the number of fault occurrences. It has been noted that none of the fault cases have a majority, while the remaining fault cases occur nearly equally.

5. Results and discussion

- **Performance evaluation and comparative analysis without syngen dataset**

A correlation is a statistical association between two features or variables. Positive correlation implies that together the features change identically i.e., an increase in the first feature results in an increase in the second feature. A negative correlation infers that the features move in the opposite direction i.e., an increase in the first feature results in a decrease in the second feature. Fig. 11 shows the correlation between the various features.

Performance measures assess the machine learning algorithm completed for a specific task. Accuracy measures the correctness of the ML algorithm predictions for the total predictions. Recall quantifies the missed correct predictions by the ML algorithm. Precision gives the confidence level in the accurate predictions made by the ML algorithms. F1 uses both precision and recall. It creates a sense of balance between

Table 2
Performance metrics different models.

Algorithm	SVC	DT	MLP	GNB	ABC
Accuracy	31.91 %	86.39 %	77.36 %	79.69 %	75.84 %
Precision	0.24	0.85	0.78	0.78	0.74
Recall	0.32	0.85	0.78	0.79	0.76
F1 score	0.32	0.85	0.77	0.76	0.74
MAE	−1.71	−0.41	−0.52	−0.5	−0.499
MSE	−5.88	−1.24	−1.49	−1.38	−1.22
RMSE	1.29	1.102	1.21	1.17	1.10

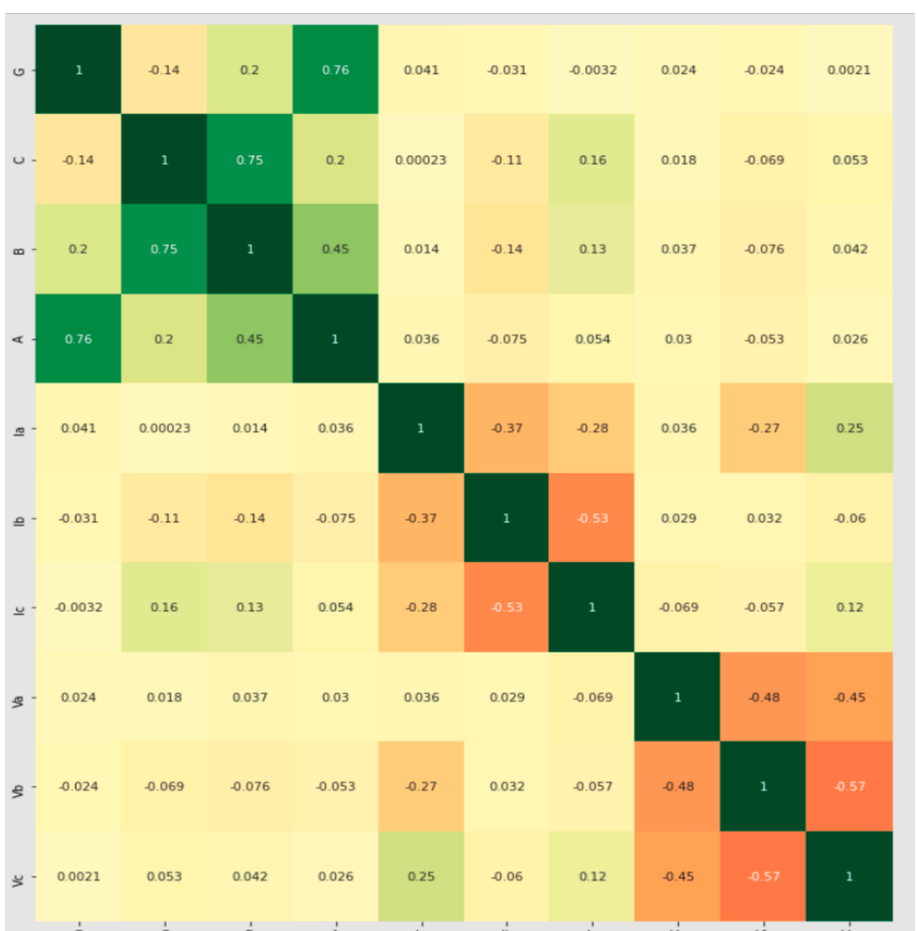

Fig. 11. Correlation between different features.

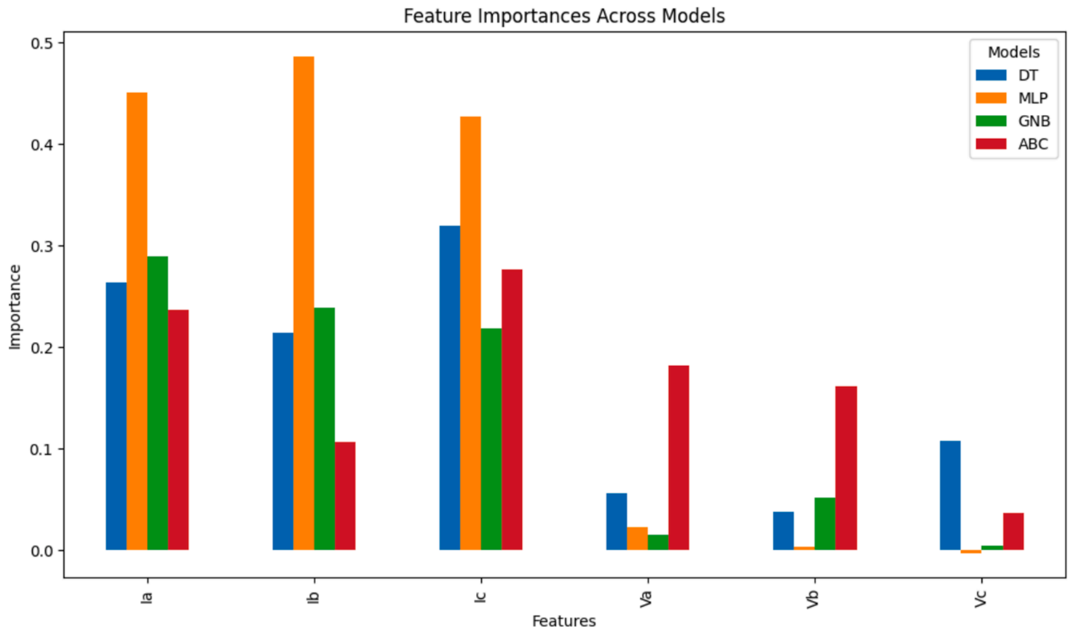

Fig. 12. Feature importance across models.

Table 3a
Hyperparameters of VAE + GAN-produced dataset.

Hyperparameter	Value
Input dimension	Equal to number of features in the dataset
Latent dimension	100
Encoder/ Decoder hidden neurons	128
Encoder/Decoder activation	ReLU
Decoder output activation	Tanh
Generator hidden neurons	128
Generator activation	ReLU
Generator output activation	Tanh
Discriminator hidden neurons	128
Discriminator activation	ReLU
Discriminator Output activation	Sigmoid
Optimizer	Adam
Learning rate	0.0002
Epochs	10,000
Batch size	32

Table 3b
Hyperparameters of GAN-produced dataset.

Hyperparameter	Value
Input dimension	Equal to number of features in the dataset
Latent dimension	100
Generator hidden neurons	128
Generator activation	ReLU
Generator output activation	Tanh
Discriminator hidden neurons	128
Discriminator activation	ReLU
Discriminator Output activation	Sigmoid
Optimizer	Adam
Learning rate	0.0002
Epochs	10,000
Batch size	32

Table 4a
Performance evaluation of SMOTE-produced dataset.

Method	Accuracy	Precision	Recall	F1-score
LR	71.5 %	0.70	0.71	0.66
ABC	100 %	1	1	1
DT	100 %	1	1	1
SVC	73.86 %	0.7	0.72	0.7
MLP	100 %	1	1	1
GNB	100 %	1	1	1

Table 4b
Performance evaluation of GAN-produced dataset.

Method	Accuracy	Precision	Recall	F1-score
LR	87.992 %	0.9	0.86	0.85
ABC	100 %	1	1	1
DT	100 %	1	1	1
SVC	82.005 %	0.74	0.72	0.72
MLP	99.05 %	0.98	0.98	0.98
GNB	100 %	1	1	1

Table 4c
Performance evaluation of KDE-produced dataset.

Method	Accuracy	Precision	Recall	F1-score
LR	47.46 %	0.23	0.35	0.27
ABC	50.118 %	0.16	0.27	0.2
DT	99.933 %	1	1	1
SVC	46.451 %	0.29	0.24	0.24
MLP	47.528 %	0.49	0.65	0.54
GNB	96.771 %	0.96	0.99	0.97

them. The F1 score implies the correct predictions of the ML algorithm and a high confidence level regarding its correctness. The scrutiny is executed on the preprocessed dataset using the algorithms. The dataset is separated into two portions – one for learning and one for evaluation. ModelSelection package in Scikit-learn runs the train_test_split class, which splits the dataset into predefined magnitudes. The dataset consists of 7861 tuples. 67 percent is considered as a learning set and 33 percent is evaluation data. The performance metrics of different ML models is shown in Table 2.

From the above Table 2, the following conclusions can be drawn for the original dataset with syn-gen:

Table 4d
Performance evaluation of PCA + GMM-produced dataset.

Method	Accuracy	Precision	Recall	F1-score
LR	52.1 %	0.28	0.4	0.32
ABC	83.417 %	0.61	0.71	0.65
DT	99.966 %	1	1	1
SVC	54.726 %	0.42	0.44	0.41
MLP	88.732 %	0.88	0.87	0.87
GNB	85.94 %	0.93	0.96	0.93

Table 4e
Performance evaluation of VAE + GAN −produced dataset.

Method	Accuracy	Precision	Recall	F1-score
LR	76.186 %	0.79	0.76	0.75
ABC	84.527 %	0.82	0.86	0.84
DT	100 %	1	1	1
SVC	81.9375	0.74	0.72	0.71
MLP	99.865 %	1	1	1
GNB	100 %	1	1	1

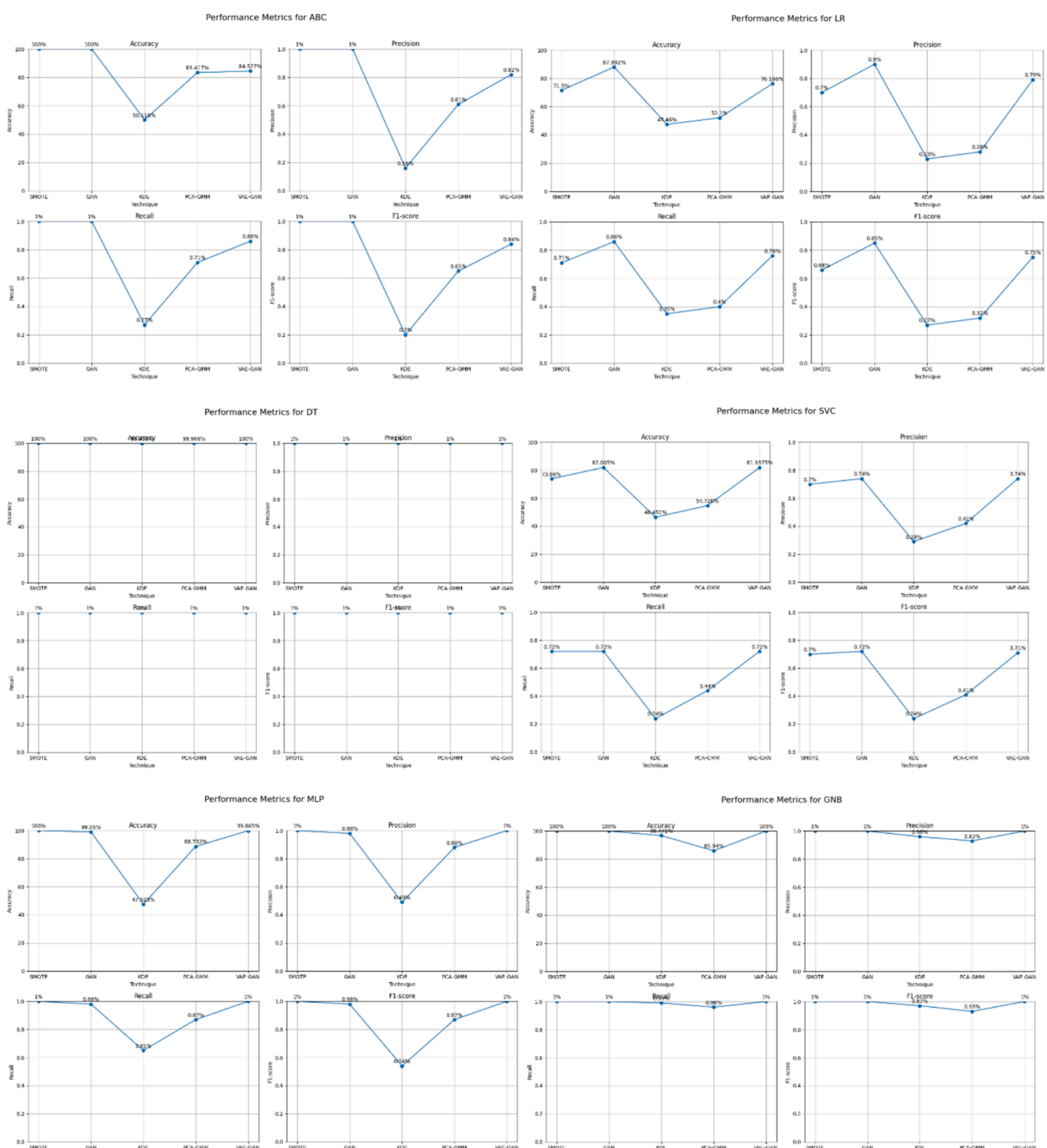

Fig. 13. Performance metrics.

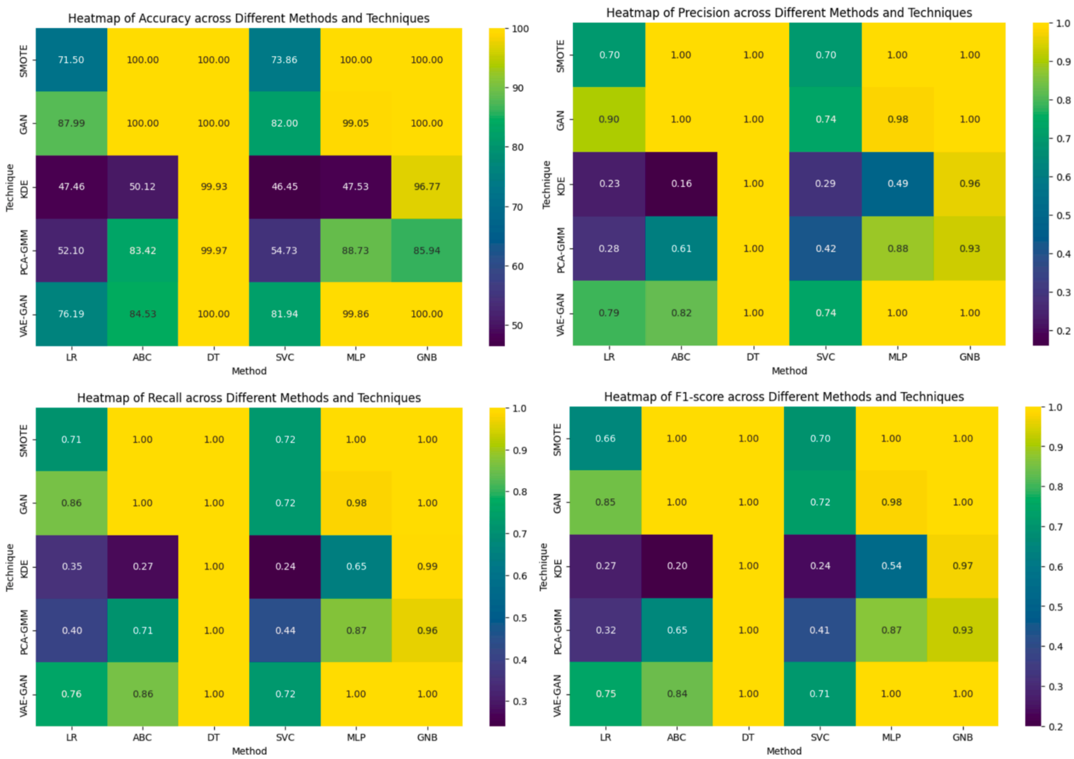

Fig. 14. Heatmap of accuracy, precision, recall, F1-score.

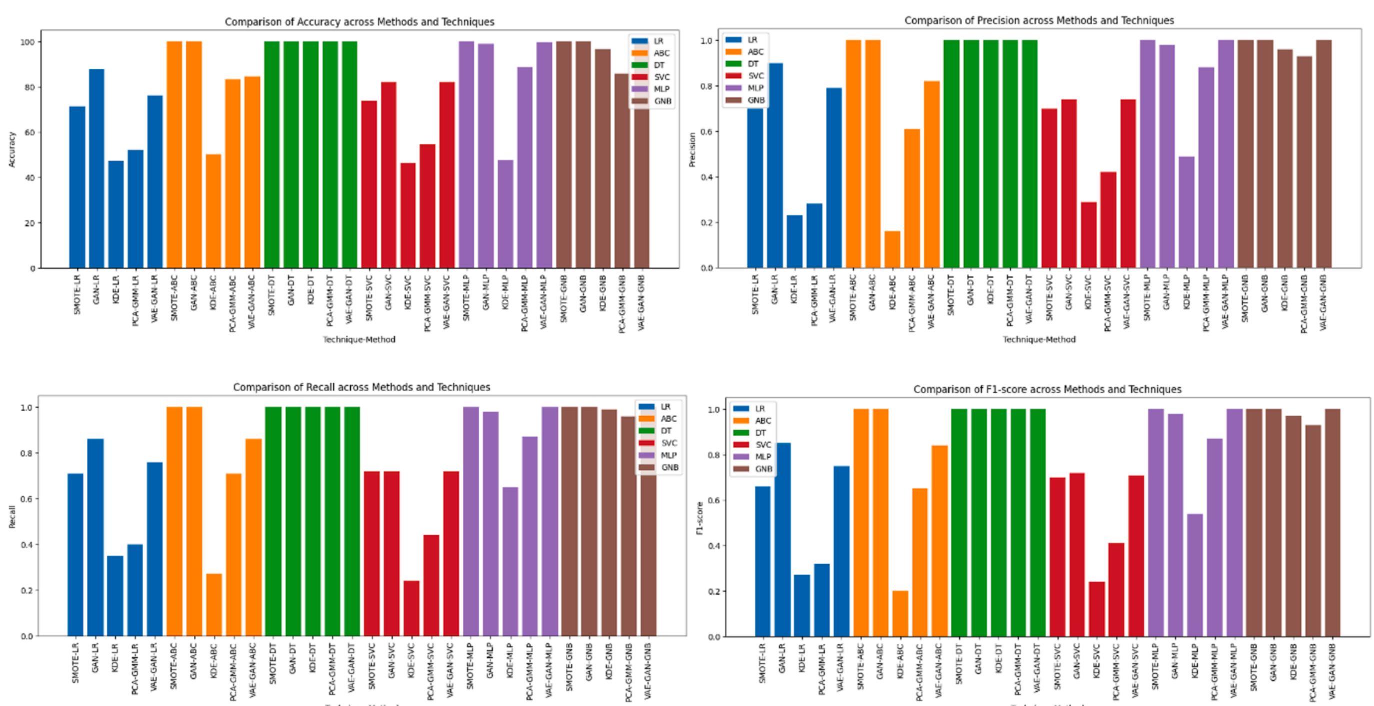

Fig. 15. Bar chart of accuracy, precision, recall, F1-Score.

1. DT gives the best results as compared with an accuracy of 86.39 %. it may be as DT is most suitable model to handle non-linear relationships between the features. SVC has the lowest accuracy of 31.91 %. it may be because the dataset may not be linearly separable.
2. MLP, GNB, and ABC show moderate performance. This means that these algorithms can handle the dataset rationally

The feature importance across ML models is given in Fig. 12.

In the next section, the syn-gen dataset will be augmented with the original dataset and the ML algorithms will be reimplemented.

- **Process of synthetic data generation and simulation**

The dataset is imported, and the data distribution is understood. The

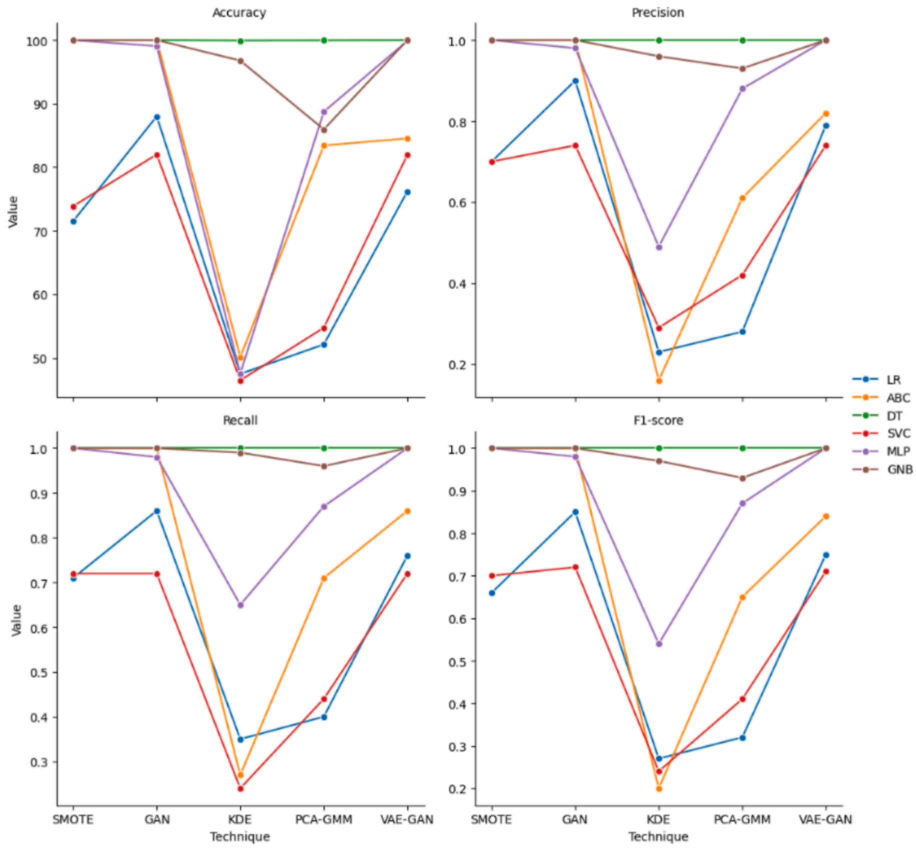

Fig. 16. Facet grid accuracy, precision, recall, F1-Score.

Table 5a
Performance evaluation of original and GAN-produced dataset.

Method	Accuracy Training	Validation	Testing	Precision Training	Validation	Testing	Recall Training	Validation	Testing	F1-score Training	Validation	Testing
LR	95.659 %	95.02 %		0.95	0.94		0.93	0.92		0.93	0.91	
ABC	92.362 %	92.09 %	91.692 %	0.78	0.78	0.78	0.85	0.85	0.85	0.81	0.81	0.8
DT	100 %	99.966 %	100 %	1	1	1	1	1	1	1	1	1
SVC	81.864 %	82.672 %	81.332 %	0.75	0.75	0.73	0.72	0.72	0.7	0.72	0.73	0.71
MLP	99.708 %	99.63 %	99.64 %	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99
GNB	100 %	100 %	100 %	1	1	1	1	1	1	1	1	1

Table 5b
Performance evaluation of original and PCA + GMM-produced dataset.

Method	Accuracy Training	Validation	Testing	Precision Training	Validation	Testing	Recall Training	Validation	Testing	F1-score Training	Validation	Testing
LR	50 %	51.48 %	50.32 %	0.16	0.16	0.17	0.27	0.27	0.27	0.2	0.2	0.21
ABC	90.579 %	91.05 %	90.279 %	0.82	0.82	0.82	0.85	0.85	0.85	0.84	0.83	0.84
DT	100 %	100 %	100 %	1	1	1	1	1	1	1	1	1
SVC	55.731 %	55.182 %	54.547 %	0.44	0.41	0.41	0.45	0.44	0.43	0.43	0.41	0.4
MLP	90.186 %	88.493 %	89.943 %	0.90	0.88	0.91	0.87	0.85	0.86	0.88	0.86	0.88
GNB	85.453 %	85.464 %	85.637 %	0.93	0.93	0.93	0.96	0.96	0.96	0.93	0.93	0.93

next step is to analyze which synthetic dataset generation technique would be suitable and the number of data points that can be generated. There should be a balance between the quantity of synthetic dataset points generated from the actual one. After the technique is selected, random sampling from the original dataset is done to generate the synthetic data points. Then, there are two options: either the synthetic dataset can be used individually, or the original and synthetic datasets can be combined to create a new dataset.

The ratio of the original and synthetic datasets combined to create a new dataset is decided through a trial-and-error method. This means that different combinations are evaluated to observe the model's performance. In some cases, the incorporation of syn-gen may lead to improved results due to better generalability and scalability. However, sometimes if the syn-gen lacks variability, the model's performance decreases due to the addition of noise. So, there is no optimum ratio to combine the original and syn-gen datasets.

So, in this paper, two test cases were created. In the first case, 7000 dataset points were synthetically generated using the syn-gen

Table 5c
Performance evaluation of original and VAE + GAN-produced dataset.

Method	Accuracy Training	Validation	Testing	Precision Training	Validation	Testing	Recall Training	Validation	Testing	F1-score Training	Validation	Testing
LR	86.187 %	86.199 %	85.699 %	0.85	0.85	0.84	0.67	0.67	0.66	0.68	0.68	0.66
ABC	100 %	100 %	99.96 %	1	1	1	1	1	1	1	1	1
DT	100 %	100 %	99.96 %	1	1	1	1	1	1	1	1	1
SVC	81.76 %	82.63 %	81.197 %	0.75	0.75	0.73	0.72	0.72	0.7	0.72	0.73	0.71
MLP	99.96 %	99.96 %	99.899 %	1	1	1	1	1	1	1	1	1
GNB	100 %	100 %	100 %	1	1	1	1	1	1	1	1	1

techniques mentioned in this paper and were used for training the ML techniques for fault classification. In the second case, the synthetic and original datasets were combined and were implemented for the learning, validating, and testing of the ML techniques for fault classification.

The performance in both cases was assessed on the accuracy, precision, recall, and F1-score. The results obtained for both cases are tabulated and visualized in the next subsection.

- **Performance evaluation and comparative analysis**

Case I: Synthetically Generated Dataset

In this case, the dataset with synthetic generation of 7000 points was used to train the ML techniques such as – LR, ABC, DT, SVC, MLP, and GNB. The performance evaluation for each of the syn-gen methods such as SMOTE, GAN, VAE-GAN, PCA-GMM, and KDE was implemented. The hyperparameters were chosen based on the domain knowledge, trial, and error method, and from the literature. The input dimension was selected to correspond to the features in the dataset, ensuring that the model could properly analyze the data. The latent dimension was determined to find a compromise between model complexity and performance. The number of neurons in the hidden layers was adjusted to a suitable amount to allow the model to learn complicated associations, and activation functions were chosen for their ability to add nonlinearity while remaining computationally efficient. The optimizer was chosen for its adaptable learning rate, which aids in the training. The batch size was set to balance memory utilization with gradient stability, and the number of epochs was set such that the model had enough time to learn and converge. The hyperparameters used for the syn-gen methods are shown in Tables 3(a) and 3(b):

The performance evaluation for each of the syn-gen methods such as SMOTE, GAN, VAE-GAN, PCA-GMM, and KDE for all the ML techniques are given below in Tables 4(a)–4(e):

Fig. 13 shows the performance metrics for LR, ABC, DT, SVC, MLP, and GNB.

Fig. 14 shows the heatmap of accuracy, precision, recall, and F1-score for the different methods and techniques.

The comparison of the accuracy, precision, recall, and F1-score across different methods and techniques is shown using the bar chart in Fig. 15.

Fig. 16 shows the facet grid accuracy, precision, recall, and F1-score across different methods and techniques.

The above tables and visualizations, the following conclusions can be drawn for the syn-gen dataset:

1. SMOTE

o ABC, DT, MLP, and GNB had 100 % accuracy, and 1 as precision, recall, and F1-score.
o SVC had accuracy of 73.86 %.
o LR performed poorly with about 71 % accuracy.

2. GAN

o ABC, DT, and GNB got 100 % accuracy.
o MLP had almost 100 % accuracy.
o The performance of LR improved as compared to SMOTE.
o For GAN, SVC had the poorest performance at accuracy of around 82 %.

3. KDE

o DT and GNB had good performance at 100 % and 96 % accuracy respectively.
o The remaining three methods performed badly with accuracy ranging from 45 % to 50 %.

4. PCA-GMM

o DT had 99 % accuracy.
o MLP, GNB, and ABC had accuracy from the range of 83 % to 85 %.
o LR and SVC had the lowest performance with an accuracy of 52 to 54 %.

5. VAE-GAN

o DT and GNB had 100 % accuracy.
o MLP had almost 100 % accuracy.
o ABC and SVC had an accuracy of around 83 to 84 %
o LR had the lowest accuracy of around 76 %.

The following generalized conclusions can be listed:

- DT and GNB constantly performed well across all syn-gen techniques.
- LR and SVC showed varying performance depending on the data generation technique, performing better with GAN and VAE-GAN compared to SMOTE, KDE, and PCA-GMM.
- GAN and VAE-GAN generally resulted in better performance for most models compared to SMOTE and KDE.
- ABC consistently performed well, particularly with SMOTE and GAN, but showed lower performance with KDE and PCA-GMM.

Case II: The original and synthetic datasets are combined and a new dataset is created. To find the ideal ratio of synthetic to original data points in the augmented datasets, a trial-and-error method was employed. It was shown that improved results were consistently produced when an equal amount of original and synthetic data points were used. This technique retained the qualities of the original data while enabling the models learn from the wide range of the synthetic data. This helped to effectively avoid both overfitting from excessive synthetic data and underfitting from inadequate augmentation. Additionally, the 1:1 ratio produced respectable performance metrics by guaranteeing dependability across different models.

The combined dataset is partitioned into three sets for the same number of dataset points, then evaluated for the various ML techniques in Tables 5(a)–5(c) and the corresponding confusion matrices in Fig. 17.

Fig. 18 shows – (a) Accuracy, (b) Validation, (c) Recall, and (d) F1-score for training, validation, and testing across the various procedures and techniques.

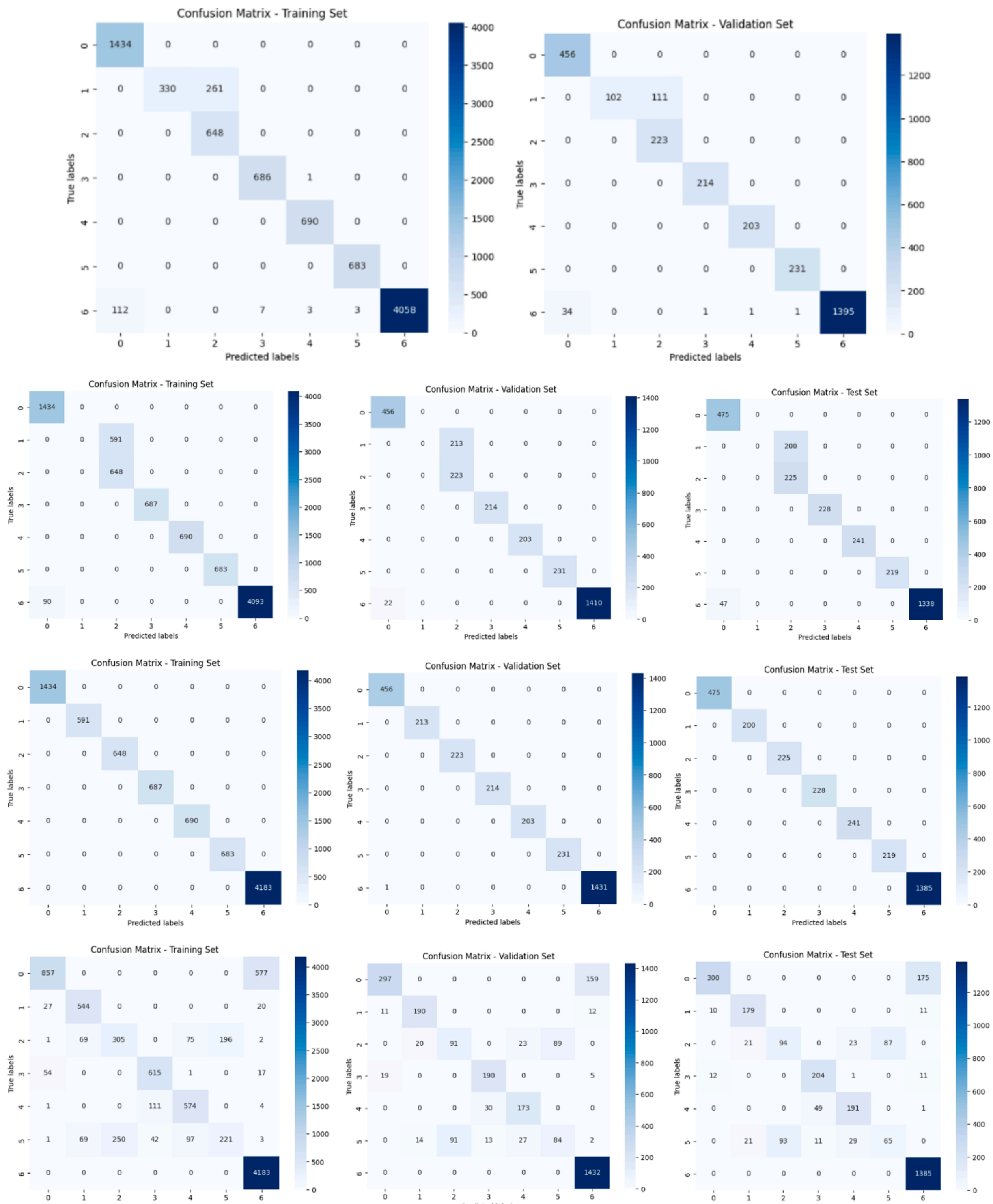

Fig. 17. **(a) 1.** Confusion matrix for LR. **(a) 2.** Confusion matrix for ABC. **(a) 3.** Confusion matrix for DT. **(a) 4.** Confusion matrix for SVC. **(a) 5.** Confusion matrix for MLP. **(a) 6.** Confusion matrix for GNB. **(b) 1.** Confusion matrix for LR. **(b) 2.** Confusion matrix for ABC. **(b) 3.** Confusion matrix for DT. **(b) 4.** Confusion matrix for SVC. **(b) 5.** Confusion matrix for MLP. **(b) 6.** Confusion matrix for GNB. **(c) 1.** Confusion matrix for LR. **(c) 2.** Confusion matrix for ABC. **(c) 3.** Confusion matrix for DT. **(c) 4.** Confusion matrix for SVC. **(c) 5.** Confusion matrix for MLP. **(c) 6.** Confusion matrix for GNB.

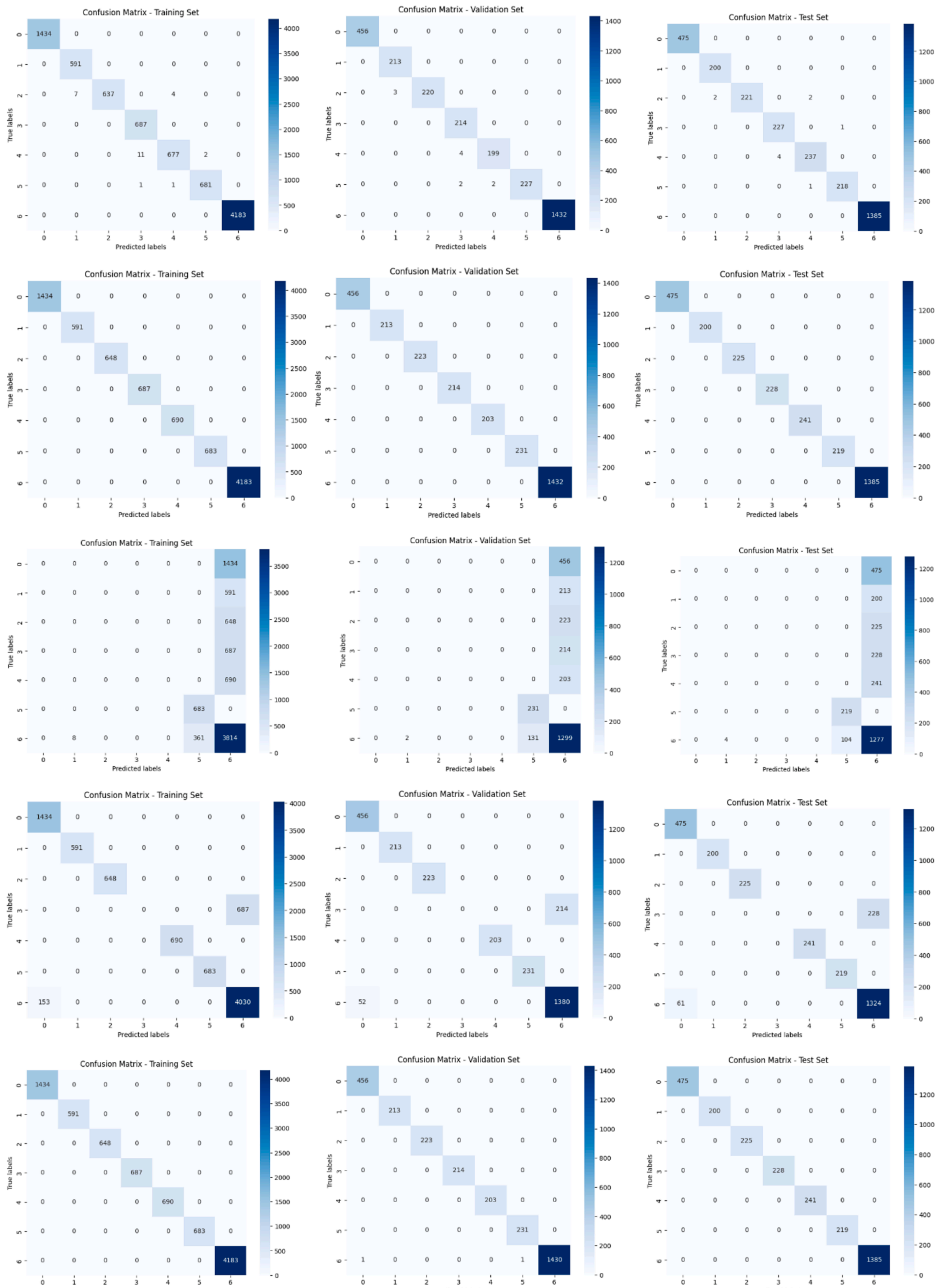

Fig. 17. (*continued*).

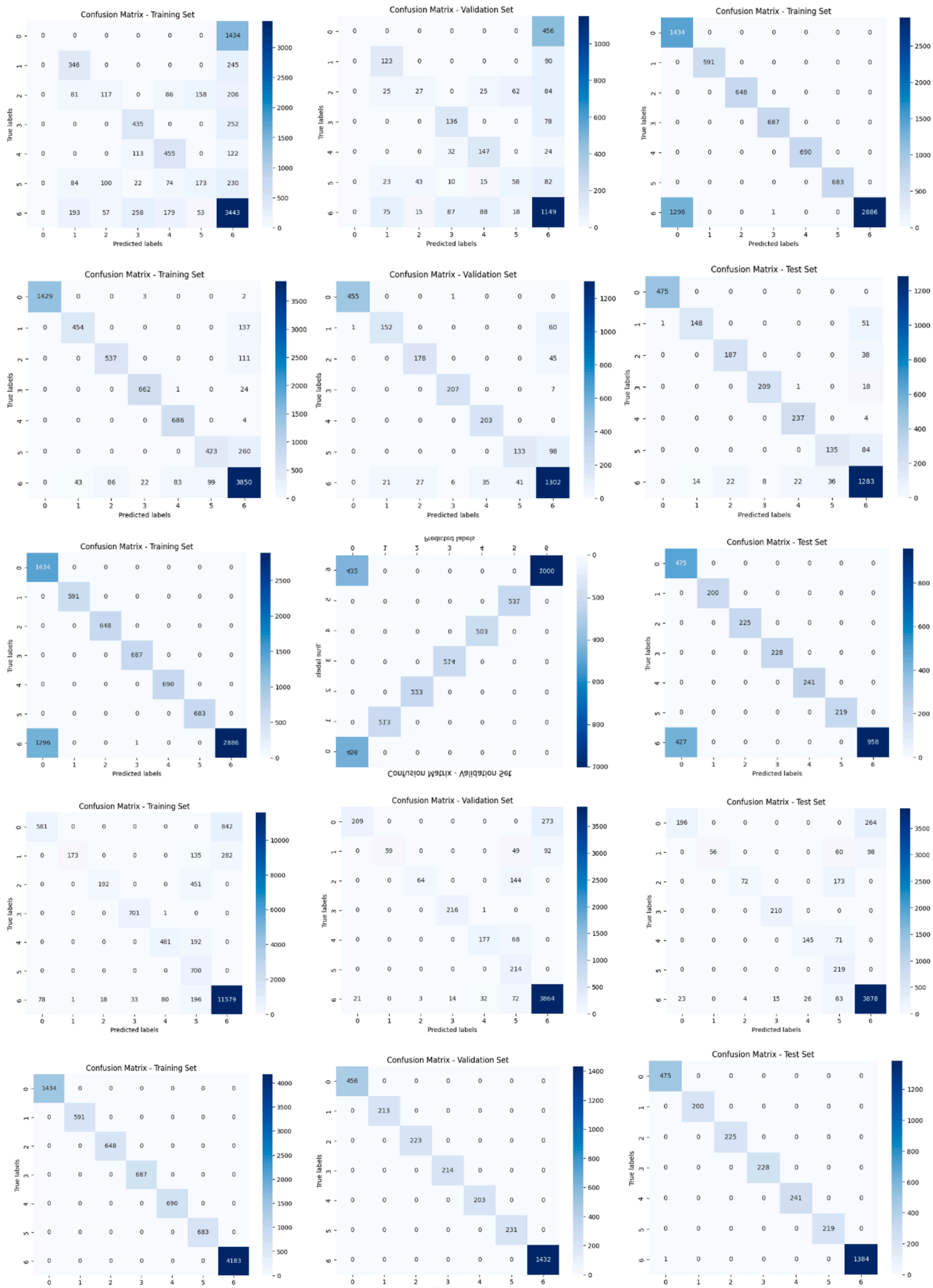

Fig. 17. (*continued*).

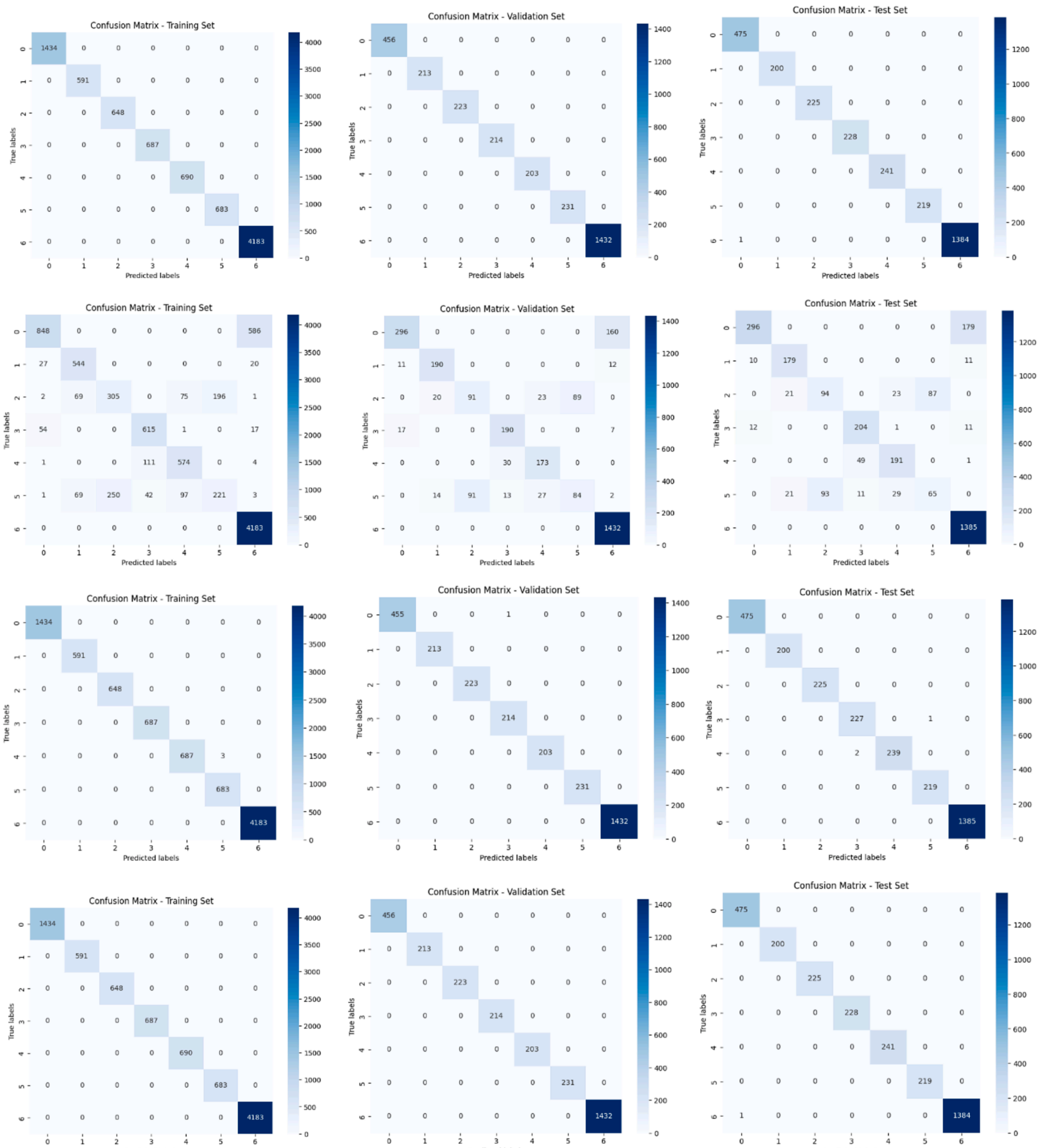

Fig. 17. (*continued*).

Fig. 19 shows the box plot for accuracy, precision, recall, and F1-score for training, validation, and testing across different methods and techniques.

The performance and generalization capacities of GAN, PCA-GMM, and VAE-GAN models may be inferred by analysing tables 2a, 2b, and 2c for these models across training, validation, and testing sets for various techniques (LR, ABC, DT, SVC, MLP, and GNB) using the augmented (original + syn-gen) dataset.

- GAN
  - o DT has 100 % accuracy and 1 as the value for precision, recall, and F1-score
  - o GNB also achieves 100 % accuracy, and 1 for precision, recall, and F1-score.
  - o MLP has the second-best performance with an accuracy of around 99.64 %, and a recall, precision, and F1-score of 0.99.
  - o LR has accuracy of 95 %, precision, recall, and an F1-score of around 0.9.
  - o ABC achieves an accuracy of around 92 %.
  - o SVC performs poorly as compared to other techniques with an accuracy of around 82 %.
- PCA-GMM
  - o The performance of DT and GNB is the same as that of GAN.
  - o The performance of LR reduced drastically and now has an accuracy of about 50 %.
  - o ABC's performance is less as compared to that of GAN, with an accuracy of around 90 %.
  - o SVC performs poorly with an accuracy of about 55 %.

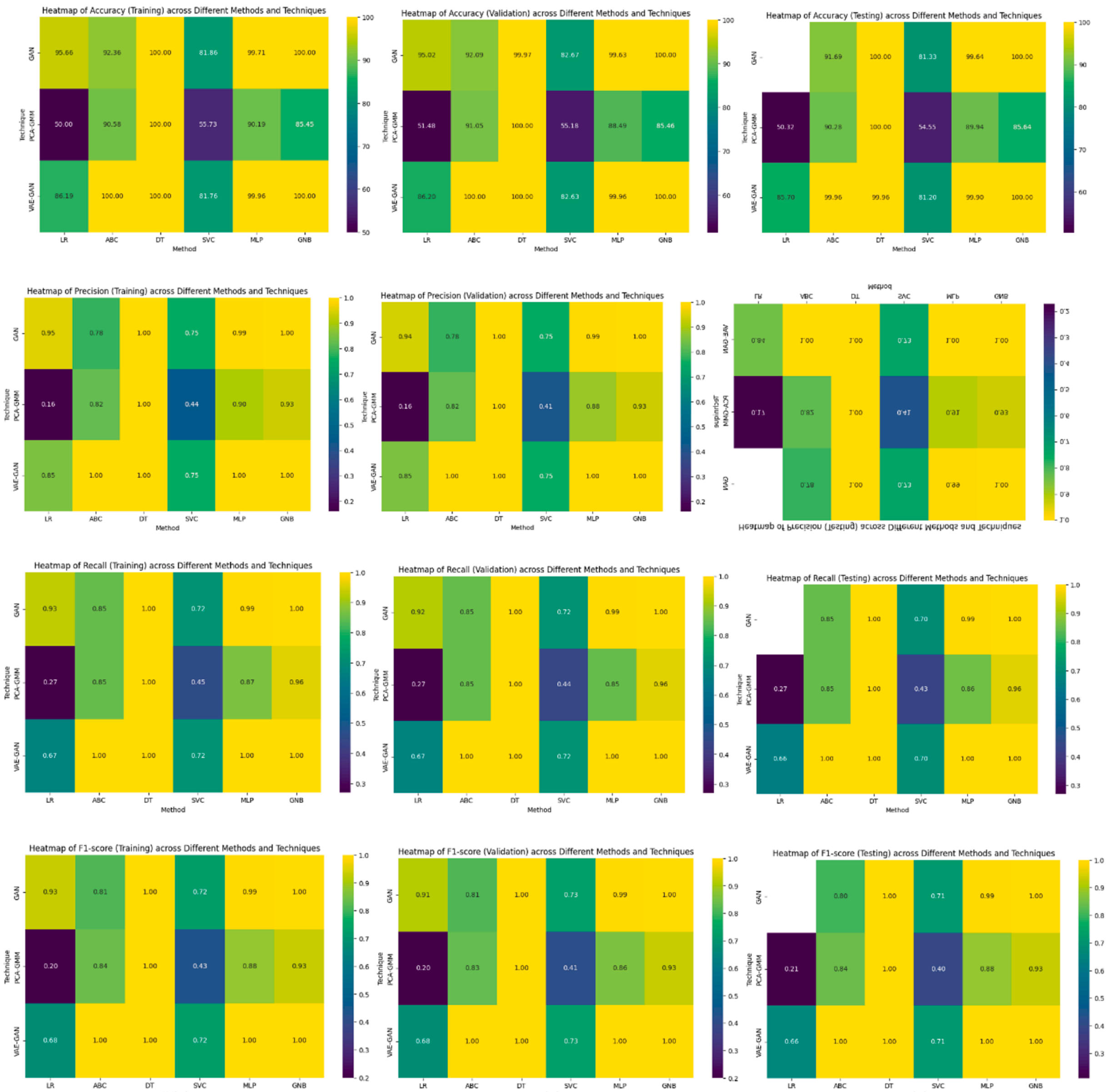

Fig. 18. (a) Accuracy, (b) Validation, (c) Recall, and (d) F1-score for training, validation, and testing.

- o Performance of MLP has reduced as compared to that with GAN to about 90 %.
- VAE-GAN
  - o DT and GNB have 100 % accuracy, precision, recall, and F1-score.
  - o The performance for LR improved with VAE-GAN with an accuracy of about 85 %.
  - o For ABC, the performance improved immensely with an accuracy of 100 %
  - o The accuracy for SVM is around 81 % which is similar to that with GAN.
  - o MLP has accuracy of almost 100 % and precision, recall, and F1-score of 1.

Perfect scores for all metrics and sets indicate that DT and GNB models could be overfitting, especially for complicated models without any mention of regularisation, like Decision Trees. However, as the training and testing performance metrics are comparable for DT and GNB, overfitting might not be the reason for the perfect scores. Mitigation strategies such as pruning, setting a minimum number of samples, performing cross-validation, or implementing ensemble methods can be explored. The performance of the dataset generated using GAN on the various ML techniques is constantly better than the other syn-gen methods. Lower performance across models (especially in LR and SVC) using PCA-GMM implies that the feature transformation may not be collecting substantial variability in the data or may not be catching all the essential information. Furthermore, VAE-GAN performs robustly with extremely high accuracy and other metrics, especially for MLP and ABC; but, for certain models, such as LR, it performs significantly worse than GAN. The performance MLP for all the syn-gen datasets is consistent but for SVC it varies depending on the syn-gen method. From this, it can be concluded that the performance of the ML algorithms is dependent on the syn-gen method.

6. Conclusion

Early defect detection and localization are essential for the dependable functioning of electrical transmission networks. Power transmission networks have benefited from the ML and DL procedures in recent years

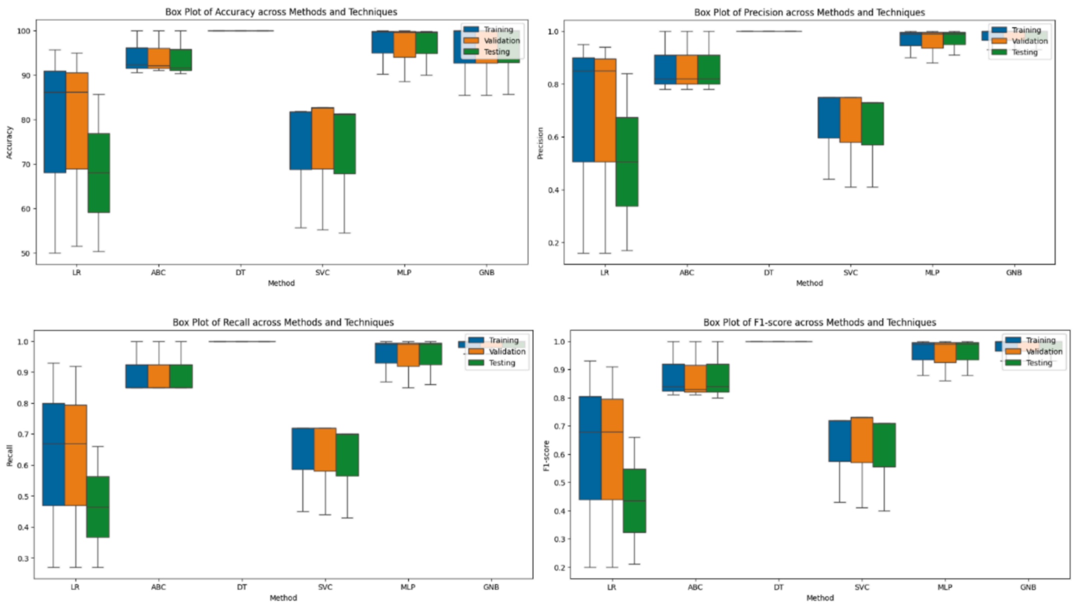

Fig. 19. Box plot for accuracy, precision, recall, and F1-score for training, validation, and testing.

in fault identification and categorization. However, the amount and competence of training data have a substantial influence on these strategies' functionality. The methods for creating synthetic data based on AI were presented in this study. Several syn-gen methods were implemented and the new dataset, thus, was created by combining this artificial data with the original dataset. The new datasets were tested on various ML techniques and their performance was analyzed.

This study compares the efficiency of several machine learning models using the original and synthetic datasets generated via various syn-gen. DT and GNB are the most effective models, with near-perfect metrics across the majority of syn-gen techniques. Their adaptability across non-linear and intricate relations in the dataset is obvious, resulting in them being appropriate choices for this dataset.

On the other hand, the effectiveness of LR and SVC changes considerably and is severely affected by the syn-gen methods implemented. These models outperform advanced syn-gen approaches such as GAN and VAE-GAN, suggesting that these methods provide more effective data distributions for linear and kernel-based classifiers. Their better accuracy and performance metrics indicate that they are more effective in creating high-quality syn-gen datasets. PCA-GMM gives satisfactory results with DT and GNB but has significantly reduced performances with LR, SVC, and MLP. The results of KDE are inconsistent performing satisfactorily for DT and GNB but poor for other models.

The comparative study emphasizes the need to choose relevant syn-gen approaches based on the required outcome and problem complexity. GAN and VAE-GAN emerge as the most adjustable methods, enhancing the generalizability of different models.

Finally, DT and GNB are suggested for applications that need high resilience across a variety of syn-gen methods, but GAN and VAE-GAN are preferable methods for creating synthetic data. This study emphasizes the need for model-specific and method-specific optimization in getting improved performance on synthetic datasets. Further investigation of statistical significance tests and real-world validations might enhance the generalizability and application of these results.

Syn-gen has certain limitations that affect its real-world applicability. The syn-gen data may sometimes not accurately represent the changeability of the real-world data. This might lead to overfitting of the ML models. If such models are used for real-world or test data, they perform poorly. This reduces the practical applicability of using the syn-gen. If the data collection process had assumptions, then the biases would be reflected in the syn-gen as well. This might lead to unfair or inaccurate outputs. The real-world data is complex. If the syn-gen is incapable of considering these complexities, then the model would also not consider them, leading to poor generalizations. Validation of the syn-gen dataset is difficult. This leads to uncertainty of whether the performance is enhanced or reduced by using syn-gen.

The syn-gen techniques look promising for the large-scale power systems. It can model critical situations without risking the grid. It can also simulate hypothetical situations, which can be beneficial for planning and development. However, the syn-gen should be based on accurate data. Any incomplete data would give unrealistic results.

In conclusion, even if the suggested techniques show a great deal of promise for classifying faults in power systems, a few crucial areas still require investigation. Future research should concentrate on improving scalability for high-dimensional datasets and large-scale power systems, facilitating integration into real-time fault monitoring systems, and resolving issues with industry adoption, such as interoperability and processing needs.

The applicability and reliability of these techniques can also be increased by using multi-objective optimization frameworks to balance classification performance, computational efficiency, and interpretability; utilizing transfer learning for broader applicability in fields like renewable energy systems; and performing robustness testing in scenarios with incomplete data or noise levels. Syn-gen can be implemented for industrial automation and predictive maintenance. They can be applied in cybersecurity to provide a safe simulation of attack scenes. It can also provide data privacy by replicating the patterns. This may be useful for research purposes. However, syn-gen may slip to record the minute patterns or timings, which can lead to a wrong training model. These avenues present encouraging chances to enhance the usefulness and influence of this study. As a future work, the deep generative models, such as diffusion models, can be implemented as they produce high-fidelity data for better representation of the real world.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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